智慧命题系统-Project Charter-齐震罡_translated.pdf

PROJECT CHARTER

Project Name: Intelligent Proposition Validation System

Project Start Date: 2026.3.30

Project End Date: 2026.6.14

Project Manager: Qi Zhengang

Project Description

Project Background: With the deepening of educational informatization, the quality of university examination paper propositions directly impacts the fairness and scientific rigor of talent cultivation. Currently, universities face challenges in the proposition process, including difficulties in monitoring repetition rates (must be below 30%), inconsistent formatting standards, and the inability of traditional plagiarism detection tools to identify semantically rewritten content. To enhance the intelligence level of teaching management, this project aims to build a validation platform integrating document parsing, deep semantic understanding, and automated reporting.

Challenges:

* Insufficient Depth of Semantic Understanding: Traditional tools rely on text matching and struggle to identify questions that have undergone synonym replacement or sentence restructuring.

* Cumbersome Format Validation: Exam paper formatting involves numerous specifications; manual verification of question numbers, point values, and fonts is inefficient and error-prone.

* Data Processing Pressure: The system needs to support OCR recognition for formulas and images and perform high-precision vector matching against a massive database of past exam papers.

Expected Goals:

* Intelligent Semantic Plagiarism Detection: Introduce deep learning models to achieve automatic marking and source tracing for paraphrased or synonym-substituted questions.

* Automated Format Correction: Implement structured parsing of DOCX documents to automatically correct formatting errors and provide visual suggestions.

* Efficient Management and Feedback: Generate a detailed "Proposition Quality Analysis Report" within 2 minutes, covering similarity rates and knowledge point distribution charts.

Measurable Organizational Value (MOV)

* Compliance Assurance: Ensure the semantic repetition rate between all validated papers and papers from the past 3 years is strictly controlled within 30%.
* Efficiency Improvement: Reduce the complete validation time for a single paper from several hours with traditional manual methods to under 2 minutes.
* Quality Optimization: Achieve an automatic detection rate of over 95% for common formatting errors (question numbers, point values, fonts).
* Decision Support: Provide a visual "Proposition Quality Analysis Report," offering quantitative teaching quality monitoring data for academic affairs departments.

Project Scope

Included (In-Scope):

* Structured parsing of DOCX documents (extraction of question stems, options, point values).
* Global formatting validation and exam paper template matching.
* OCR recognition and semantic conversion of formulas and images.
* Deep learning-based semantic plagiarism detection and tag-based management for questions.
* Automated generation of multi-dimensional validation reports.
* Construction and persistent storage of a database of past exam papers.

Excluded (Out-of-Scope):

* Automatic typesetting or layout generation for exam papers (only correction suggestions provided).
* Direct parsing of document formats other than DOCX (e.g., PDF, LaTeX).
* Non-proposition-related academic functions such as exam registration and grade management.

Project Budget

* Labor Input: Total estimated effort: 324 person-hours.
* Technical Costs: Includes costs for deploying a local GPU computing platform, API call quotas for large language models, and server maintenance.
* Data Costs: Costs associated with collecting, cleaning, and storing historical exam paper data.

Core Success Criteria for Projects

* Functional Completeness: The system successfully implements the four core modules: document parsing, format validation, semantic plagiarism detection, and report generation, with no major defects in integration testing.

* Performance Indicators:

* Format error detection rate ≥ 95%

* Semantic rewrite question identification rate ≥ 90%

* Consistency rate of manual review for duplication judgments ≥ 90%

* Time from upload to report generation for a single paper (based on 10 pages/30 questions) ≤ 2 minutes

* User Acceptance: After trial use by at least 3 proposition teachers (including the requirement owner, Associate Professor Li Bing), the satisfaction score for system usability and result accuracy ≥ 4.5 points (out of 5).

* Repetition Rate Control Effect: Papers modified with system assistance achieve a repetition rate ≤ 30% compared to papers from the past 3 years (subject to verification by the academic affairs department).

Core Methods of Project

* Technical Approach: Use Python for backend development, utilize vectorization models (e.g., BERT/RoBERTa) to extract semantic vectors, integrate OCR engines for non-text content, and employ large language models (LLMs) for final semantic logic determination.

* Management Method: Adopt phased milestone management, ensuring parallel progress of functional modules through modular development.

Resources Required for Project

* Human Resources: 4 student members (1 project manager, 3 developers). The requirement owner (Associate Professor Li Bing) provides domain support. The academic affairs department provides baseline data samples.

* Software Resources: Python 3.10+, PyTorch, Transformers library, PaddleOCR, FastAPI, Vue.js, PostgreSQL, GitLab (code repository), Jira (project management).

* Hardware Resources:

* Development & Test Server: CPU 8 cores+, RAM 32GB+, GPU (NVIDIA RTX 3060 or above, VRAM 12GB+) for model inference.

* Storage Space: 500GB+ (for storing the past exam paper database and logs).

* Data Resources: Complete past exam papers from the past 3 years for the same courses at Tongji University (at least 5 courses, 10 papers per course) as the baseline database; exam paper format specification documents (provided by the Academic Affairs Office).

Project Management

* Project Management Method: Adopt the Agile Scrum framework with two-week Sprints. Use Jira for task tracking and burn-down chart management. Hold 15-minute daily stand-up meetings (synchronizing progress and obstacles) weekly. Hold review

and retrospective meetings at the end of each Sprint.

* Communication Plan:

* Internal Communication: QQ group for instant messaging + offline weekly meeting every Friday afternoon (30 minutes).

* Communication with Stakeholders: Submit progress reports at the end of each Sprint; conduct formal demonstrations at each milestone node (4/26, 5/24, 6/14).

* Document Management: All technical documents, design diagrams, and meeting minutes are stored uniformly in the GitLab Wiki; code comments follow the Google Python Style Guide.

* Quality Assurance: Each module must pass unit tests and code review before merging; integration testing is organized by the project manager. Bug fix priorities are categorized by severity (P0 Blocking → P1 Critical → P2 General → P3 Suggestion).

* Risk and Change Control: Major scope changes require written agreement from all members and stakeholders; risk response measures are recorded in the risk register and updated monthly.

Project Gantt Chart

(Chart would be inserted here, translation maintains placeholder)

Main Responsibilities of Members

Name	Identity	Responsibility	Contact Information
Qi Zhengang	Project Manager	1. Overall project planning, progress monitoring, and risk management. 2. Authoring core documents such as Project Charter, WBS, RBS. 3. Development of pattern matching algorithm (based on tag similarity). 4. Development of user system and validation results database. 5. Organizing testing, user manual writing, and final delivery.	1578747475@qq.com

Name	Identity	Responsibility	Contact Information
Lü Kuichen	Project Member	1. Implementation of Word document parsing and automated question segmentation. 2. Implementation of global format validation algorithm. 3. Module integration testing (collaborating with Yi Mingjun). 4. Assisting with bug fixes.	2419485769@qq.com
Yi Mingjun	Project Member	1. Development environment setup and code standard formulation. 2. OCR model integration (formula and image to text conversion). 3. Development of large language model (LLM) semantic deep matching interface. 4. Deployment of local computing platform and model inference acceleration. 5. Module integration testing (collaborating with Lü Kuichen).	1664382962@qq.com

Name	Identity	Responsibility	Contact Information
Zhao Zeyuan	Project Member	1. Implementation of exam paper template matching and missing check functions. 2. Development of automatic question tagging model. 3. Database structure design and data import for the question database (past questions). 4. Assisting with user system testing.	2556123498@qq.com